

Day 03- NETWORK INTERFACES

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Day- 03

1. What is Network Interface Card?

A Network Interface Card (NIC) is a hardware component that allows a computer to connect to a network. It can provide Ethernet or Wi-Fi connectivity.

2. What is Ethernet?

Ethernet is a networking technology used to connect devices over a wired Local Area Network (LAN).

3. What is RJ-45?

RJ-45 is the common connector used with Ethernet network cables such as Cat5e and Cat6.

4. What does UTP stand for?

- UTP stand for Unshielded Twisted Pair.
- It is a type of copper network cable containing twisted pairs of wires.

5. Name three common Ethernet cable categories.

Three common categories are:

- Cat5e
- Cat6
- Cat6a

6. What is a straight-through cable?

A straight-through cable has the same wiring standard on both ends. It is traditionally used to connect different types of devices, such as a computer to a switch.

7. What is a crossover cable?

A crossover cable has different transmit and receive wire arrangements on each end. It was traditionally used to connect similar devices, such as computer-to-computer.

8. Explain Auto MDI-X?

Auto MDI-X automatically detects the transmit and receive connections and adjusts them when necessary. Therefore modern devices can usually work without manually choosing a straight-through or crossover cable.

9. Difference between copper and fibre-optic networking?

Copper	Fiber-optic
• Uses electrical signals • Usually cheaper • Shorter transmission distance • More affected by electromagnetic interference • Easier to install	• Uses light signals • Usually more expensive • Longer transmission distance • Resistant to electromagnetic interference • Installation can be more specialised

10. What is multimode fibre?

Multimode fiber (MMF) is an optical fibre that allows multiple light paths/modes to travel through the fiber. It is commonly used for shorter-distance network connections.

11. What is signal-mode fiber?

Signal-mode fiber (SMF) allows essentially one mode of light to travel through the fiber. It is suitable for long-distance and high-bandwidth communication.

12. Two advantages of fiber-optic networking?

1. High bandwidth and fast data transmission.
2. Long transmission distance with low electromagnetic interference.

13. Why is understanding network interfaces important for cybersecurity?

Understanding network interfaces is important because cybersecurity professionals need to know how devices connect to networks, how IP addresses are assigned, and how network traffic enters and leaves a system. This helps with tasks such as network monitoring, traffic Analysis, troubleshooting, and detecting suspicious activity.

14. Identify your active network adapter.

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Command Prompt

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
Physical Address. . . . . : 10-5F-AD-57-A5-53
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . : Yes
IPv6 Address. . . . . : 2409:40d4:16f:c956:15b0:ca69:b93b:3262(Preferred)
Temporary IPv6 Address. . . . . : 2409:40d4:16f:c956:9cb:9ea4:b91f:b00e(Preferred)
Link-local IPv6 Address . . . . . : fe80::d455:e572:89dc:1154%9(Preferred)
IPv4 Address. . . . . : 10.44.64.107(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 09 September 2026 15:41:30
Lease Expires . . . . . : 09 September 2026 16:41:29
Default Gateway . . . . . : fe80::2052:2fff:fe5f:abc9%9
                          10.44.64.177
DHCP Server . . . . . : 10.44.64.177
DHCPv6 IAID . . . . . : 118513581
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-0E-51-AA-28-C5-C8-5B-29-66
DNS Servers . . . . . : 10.44.64.177
                          2409:40d4:16f:c956::3b
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
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15. Write your IPv4 address, default gateway, and network adapter name.

- IPv4:- 10.44.64.107
- Default gateway:- fe80::2052:2fff:fe5f:abc9%9
- Network adapter:- Wi-Fi

Ipconfig /all

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Command Prompt
C:\Users\AnshS>ipconfig /all

Windows IP Configuration

Host Name . . . . . : 
Primary Dns Suffix . . . . . : 
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek Gaming GbE Family Controller
Physical Address. . . . . : 28-C5-C8-5B-29-66
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Unknown adapter McAfee VPN:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-93-2E-0D-C9
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:
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Command Prompt

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 10-5F-AD-57-A5-54
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 12-5F-AD-57-A5-53
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . : 
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet1
Physical Address. . . . . : 00-50-56-C0-00-01
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::507b:6f77:6d66:bb10%8(Preferred)
IPv4 Address. . . . . : 192.168.238.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 09 September 2026 15:40:58
Lease Expires . . . . . : 09 September 2026 16:10:49
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Command Prompt
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.238.254
DHCPv6 IAID . . . . . : 805326934
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-0E-51-AA-28-C5-C8-5B-29-66
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

Connection-specific DNS Suffix . :
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8
Physical Address. . . . . : 00-50-56-C0-00-08
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::862:bb1c:99c0:825d%17(Preferred)
IPv4 Address. . . . . : 192.168.64.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 09 September 2026 15:40:50
Lease Expires . . . . . : 09 September 2026 16:10:49
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.64.254
DHCPv6 IAID . . . . . : 822104150
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-0E-51-AA-28-C5-C8-5B-29-66
Primary WINS Server . . . . . : 192.168.64.2
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
Physical Address. . . . . : 10-5F-AD-57-A5-53
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Command Prompt
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
Physical Address. . . . . : 10-5F-AD-57-A5-53
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2409:40d4:16f:c956:15b0:ca69:b93b:3262(Preferred)
Temporary IPv6 Address. . . . . : 2409:40d4:16f:c956:9cb:9ea4:b91f:b00e(Preferred)
Link-local IPv6 Address . . . . . : fe80::d455:e572:89dc:1154%9(Preferred)
IPv4 Address. . . . . : 10.44.64.107(Preferred)
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DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-0E-51-AA-28-C5-C8-5B-29-66
DNS Servers . . . . . : 10.44.64.177
                          2409:40d4:16f:c956::3b
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 10-5F-AD-57-A5-57
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